

Jonathan Lindbloom

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29 N. Main Street, Hanover, NH 03755

EDUCATION

Dartmouth College, Hanover, NH

Ph.D. in Mathematics (advised by Anne Gelb)

A.M. in Mathematics

Expected Jun. 2026

Awarded Jun. 2023

Southern Methodist University, Dallas, TX

B.S. in Mathematics (departmental distinction)

B.B.A. in Finance

Aug. 2017 – May 2021

RESEARCH INTERESTS

Inverse problems, uncertainty quantification, computational science, imaging,
numerical linear algebra, surrogate modeling, signal processing

PUBLICATIONS

3. **Lindbloom, J.**, GLAUBITZ, J., AND GELB, A. (2025).
Efficient sparsity-promoting MAP estimation for Bayesian linear inverse problems.
Inverse Problems, 41(2), 025001.
[DOI:10.1088/1361-6420/ada17f](https://doi.org/10.1088/1361-6420/ada17f)
2. GREEN, D., **Lindbloom, J.**, AND GELB, A. (2025).
Complex-Valued Signal Recovery Using a Generalized Bayesian LASSO,
SIAM/ASA Journal on Uncertainty Quantification, 13(2), 831-861.
[DOI:10.1137/24M1644778](https://doi.org/10.1137/24M1644778)
1. GREEN, D., **Lindbloom, J.**, AND GELB, A. (2024).
Empirical Bayesian inference for complex-valued signals using support-informed priors,
AIMS Applied Mathematics for Modern Challenges, 2(4).
[DOI:10.3934/ammc.2024020](https://doi.org/10.3934/ammc.2024020)

PREPRINTS

1. **Lindbloom, J.**, PASHA, M., GLAUBITZ, J., AND MARZOUK, Y. (2025).
Priorconditioned Sparsity-Promoting Projection Methods for Deterministic and Bayesian Linear
Inverse Problems.
[arXiv:2505.01827](https://arxiv.org/abs/2505.01827).

CONFERENCE PROCEEDINGS

1. GREEN, D., **Lindbloom, J.**, AND GELB., A. (2024).
Complex-Valued Image Recovery from Multiple Measurements,
2024 IEEE Conference on Computational Imaging Using Synthetic Apertures (CISA).
[DOI:10.1109/CISA60639.2024.10576266](https://doi.org/10.1109/CISA60639.2024.10576266)

TALKS

24. *Preconditioning techniques for large-scale sparsity-promoting inverse problems* *Apr. 2025*
Dartmouth Applied & Computational Mathematics Seminar (Hanover, NH)
(slides)
23. *Preconditioning techniques for large-scale sparsity-promoting inverse problems* *Apr. 2025*
East Coast Optimization Meeting (Arlington, VA)
(slides)
22. *Preconditioning techniques for large-scale sparsity-promoting inverse problems* *Mar. 2025*
50th Annual New York State Regional Graduate Mathematics Conference (Syracuse, NY)
21. *Preconditioning techniques for large-scale sparsity-promoting inverse problems* *Mar. 2025*
SIAM Conference on Computational Science and Engineering (Fort Worth, TX)
(slides)
20. *Cauchy interlacing and generalized Ostrowski theorems* *Jan. 2025*
Dartmouth Mathematics Graduate Student Seminar (Hanover, NH)

19. *Priorconditioned generalized Krylov subspaces for sparsity-promoting inversion* Dec. 2024
Virginia Tech Applied Numerical Analysis Seminar (Blacksburg, VA)
18. *Priorconditioned generalized Krylov subspaces for sparsity-promoting inversion* Oct. 2024
Dartmouth Mathematics Graduate Student Seminar (Hanover, NH)
17. *The conjugate gradient method: a top 10 algorithm of the 20th century* May 2024
Dartmouth Mathematics Graduate Student Seminar (Hanover, NH)
(photo)
16. *Generalized sparsity promoting solvers and UQ for Bayesian inverse problems* Feb. 2024
CMCC Foundation Webinar (Bologna, IT)
(slides) (video)
15. *Generalized sparsity-promoting solvers and samplers for hierarchical inverse problems* Feb. 2024
SIAM Conference on Uncertainty Quantification (Trieste, IT)
(slides)
14. *A brief introduction to matrix-free and randomized matrix computations* Jan. 2024
Dartmouth Mathematics Graduate Student Seminar (Hanover, NH)
13. *Oblique projections and low-rank structure in inverse problems* Sept. 2023
Dartmouth Mathematics Graduate Student Seminar (Hanover, NH)
(slides)
12. *Computational strategies for Bayesian inversion with sparsity priors* Mar. 2023
SIAM Conference on Computational Science and Engineering (Amsterdam, NL)
(slides)
11. *Hierarchical Bayesian inverse problems* Jan. 2023
Dartmouth Mathematics Graduate Student Seminar (Hanover, NH)
(slides)
10. *Generalized hybrid solvers for sparsity-promoting Bayesian inverse problems* Nov. 2022
Dartmouth Advancement Examination (Hanover, NH)
(slides)
9. *Generalized hybrid solvers for sparsity-promoting Bayesian inverse problems* Nov. 2022
Sea Ice Modeling and Data Assimilation MURI Annual Meeting (Hanover, NH)
(slides) (video)
8. *Surrogate modeling for high explosives diameter effect calculations* Aug. 2022
LANL Theoretical Division Lightning Talks (Los Alamos, NM)
7. *Multiplicative denoising with UQ for synthetic aperture radar* Apr. 2022
SIAM Conference on Uncertainty Quantification (Atlanta, GA)
(slides)
6. *Towards UQ for synthetic aperture radar despeckling* Oct. 2021
Sea Ice Modeling and Data Assimilation MURI Annual Meeting (Hanover, NH)
(slides) (video)
5. *Uncertainty quantification in high explosives equations of state* Aug. 2021
LANL Computational Physics Summer Workshop (virtual)
(slides)
4. *A Bayesian Gaussian process model for COVID-19* Mar. 2021
SMU Spring Research Days (virtual)
3. *Bayesian perspectives on COVID-19 in Texas* Oct. 2020
SMU Fall 2020 Summer Research Symposium (virtual)
(video)
2. *Uncertainty quantification of ice fog events* Aug. 2020
SMU Grand Challenges Scholars Research Symposium (Dallas, TX)
1. *Uncertainty quantification of ice fog events* Aug. 2020
Dartmouth ICE REU (virtual)

POSTERS	7. <i>Efficient Gaussian sampling for sparsity-promoting Bayesian inversion</i> New England Numerical Analysis Day (Hanover, NH)	<i>Jun. 2024</i>
	6. <i>Efficient Gaussian sampling for sparsity-promoting Bayesian inversion</i> Dartmouth Guarini Graduate Poster Session (Hanover, NH)	<i>Apr. 2024</i>
	5. <i>Sparsity-promoting multiplicative denoising via block coordinate descent</i> SIAM Power of Diversity UQ24 (Trieste, IT) (poster)	<i>Feb. 2024</i>
	4. <i>Sparsity-promoting multiplicative denoising via block coordinate descent</i> AFOSR Electromagnetics Annual Portfolio Review (Washington, D.C.) (poster)	<i>Jan. 2024</i>
	3. <i>Uncertainty quantification of ice fog events</i> Joint Mathematics Meeting (virtual) (poster)	<i>Jan. 2021</i>
	2. <i>Bayesian SIR Models for COVID-19 in Texas</i> NSF Student Conference on COVID-19 Modelling (virtual) (poster)	<i>Jan. 2021</i>
	1. <i>Light beams at nonlinear interfaces</i> SIAM TX-LA Annual Section Meeting (Dallas, TX) (poster)	<i>Nov. 2019</i>

LAB EXPERIENCE	Los Alamos National Laboratory , Theoretical Division (T-1)	
	<i>Graduate research assistant</i> (mentor: Jeff Leiding)	<i>Jun. 2024 – Sept. 2024</i> <i>Jun. 2022 – Sept. 2022</i>
	<i>Computational physics student summer workshop</i>	<i>Summer 2021</i>

TEACHING EXPERIENCE	Dartmouth College , Hanover, NH	
	<i>Instructor</i>	
	• Math 56: Computational and Experimental Mathematics	<i>Winter 2025</i>
	• Math 3: Calculus	<i>Fall 2023</i>
	<i>Graduate teaching assistant</i>	
	• Math 22: Linear Algebra and Applications	<i>Spring 2023</i>
	• Math 56: Computational and Experimental Mathematics	<i>Winter 2023</i>
	• Math 3: Calculus	<i>Winter 2023</i>
	• Math 3: Calculus	<i>Winter 2022</i>
	• Math 3: Calculus	<i>Fall 2021</i>

OUTREACH	Dartmouth Mathematics Directed Reading Program , Hanover, NH	
	<i>Graduate student mentor</i>	
	The goal of this program is to enable undergraduate students to study mathematics at a deeper level than can be done in the classroom, to increase diversity in mathematics by involving undergraduates from various backgrounds, and to foster a supportive environment for students seeking to go into mathematics.	
	• Randomized Numerical Linear Algebra (student: Michael Bond)	<i>Spring 2025</i>
	• MCMC and Bayesian computation (students: Michael Bond and Vadin Thadhani)	<i>Spring 2024</i>
	• Computational Measure Transport (students: William O'Brien and Paul Chirkov)	<i>Winter 2024</i>
	• MCMC and Bayesian computation (student: Daniel Carstensen)	<i>Spring 2023</i>
	• Bayesian modelling and computation (student: Ivy Yan)	<i>Spring 2022</i>
	• Nonlinear dynamics, chaos, and ergodicity (student: Andrew White)	<i>Fall 2022</i>

AWARDS	• SIAM Student Travel Award	<i>Mar. 2025</i>
	• SIAM Student Travel Award	<i>Apr. 2022</i>
	• Outstanding Graduate Student Teacher, Dartmouth College	<i>2022 – 2023</i>
	• Charles J. Pipes Merit Award in Mathematics, Southern Methodist University	<i>Spring 2021</i>
	• Florence Nightingale Prize for Data Visualization, ISI	<i>Spring 2021</i>

PROFESSIONAL SERVICE	<i>Seminar organizer</i>	
	• Dartmouth College Applied & Computational Mathematics Seminar	<i>Fall 2023 – Present</i>
	<i>Peer reviewer:</i> SIAM Journal on Scientific Computing, Inverse Problems, IEEE Transactions on Network Science and Engineering, Applied Mathematics for Modern Challenges	